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Influence of lipid bilayer head group polarity on micelle behavior and surfactant transfer: a molecular dynamics simulation study†

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Molecular dynamics simulations with umbrella sampling were conducted to investigate the behavior of micelles on polar and non-polar lipid bilayer surfaces and surfactant transfer from the SLE2S micelle to the bilayers. Two-dimensional potential of mean force (PMF) profiles reveal that the micelle avoids interaction with the polar 1,2-dimyristoyl-*sn-glycero*-3-phosphocholine (DMPC) bilayer, maintaining its spherical shape. In contrast, it adheres tightly to the non-polar CER NS 24 : 0 ceramide (Cer240) bilayer, exhibiting an upper hemispherical shape. Electrostatic interactions govern the micelle behavior at the DMPC bilayer whereas both electrostatic and van der Waals interactions influence the shape and adhesion to the Cer240 bilayer. One-dimensional PMF curves suggest that surfactant transfer is unlikely in the DMPC bilayer at either 313.15 K or 350.15 K due to an energy barrier induced by interfacial water. However, transfer can occur in the Cer240 bilayer at 350.15 K, facilitated by the loose packing of head groups.

1. Introduction

Surfactants, as amphiphilic molecules, play a crucial role in pharmaceutical, food, and cosmetics industries, serving as drug delivery carriers, emulsifying agents, or penetration enhancers.[1–3] Owing to their amphiphilic nature, surfactants self-assemble into micelles, facilitating the delivery of hydrophobic compounds across cell membranes or through the skin.[4,5] For surfactant micelles with charged or polar head groups, electrostatic interactions with cell membranes strongly influence the delivery of therapeutic agents into cells.[6,7] Therefore, comprehensively understanding the interactions of surfactant micelles with cell membranes is essential for developing novel drug delivery systems (DDSs) based on surfactant micelles.[8–10] Sodium lauryl ether sulfates (SLESs, also known as sodium laureth sulfates) represent a class of anionic surfactants that are widely found in personal care products (Fig. 1a). Due to their lower toxicity and cost-effective production, SLESs are preferred over sodium dodecyl sulfate (SDS).[11–14] Micelles formed from SLES surfactants exhibit excellent stabilizing and emulsifying activities toward hydrophobic organic molecules.[15]

Molecular dynamics (MD) simulations represent a powerful computational tool for exploring complex biological systems, such as cell membranes, at atomic and molecular levels.[16–18] When combined with enhanced sampling methods, such as umbrella sampling (US) or metadynamics, MD simulations can provide one-dimensional (1D) or two-dimensional (2D) free energy profiles, enabling the characterization of static and dynamic properties of complex systems.[19,20] Although numerous MD simulation studies have assessed the structural properties of surfactant micelles in the aqueous phase or at the air/water interface,[14,21–27] MD simulations exploring the interaction of surfactant micelles with cell membranes remain limited.[9] In our previous study, a series of MD simulations revealed distinct structural behaviors of SLE2S micelles on DMPC and ceramide lipid bilayers, including surfactant transfers from the micelle to the ceramide lipid bilayer.[18] However, a more detailed investigation is required to fully understand the energetics and mechanisms underlying these phenomena. In this study, we performed all-atom MD simulations combined with the US method to examine the energetic and structural properties of SLES micelles interacting with two model membranes: 2-dimyristoyl-*sn-glycero*-3-phosphocholine (DMPC) and CER NS 24 : 0 ceramide (Cer240) bilayers (Fig. 1b and c). DMPC lipid molecules, a type of phospholipid, are commonly used to study cell membranes,[28] while Cer240 lipids are representative ceramides that serve as key components of the stratum corneum in the skin.[29] We selected DMPC and Cer240 lipids as bilayer components due to their distinct head group properties: DMPC

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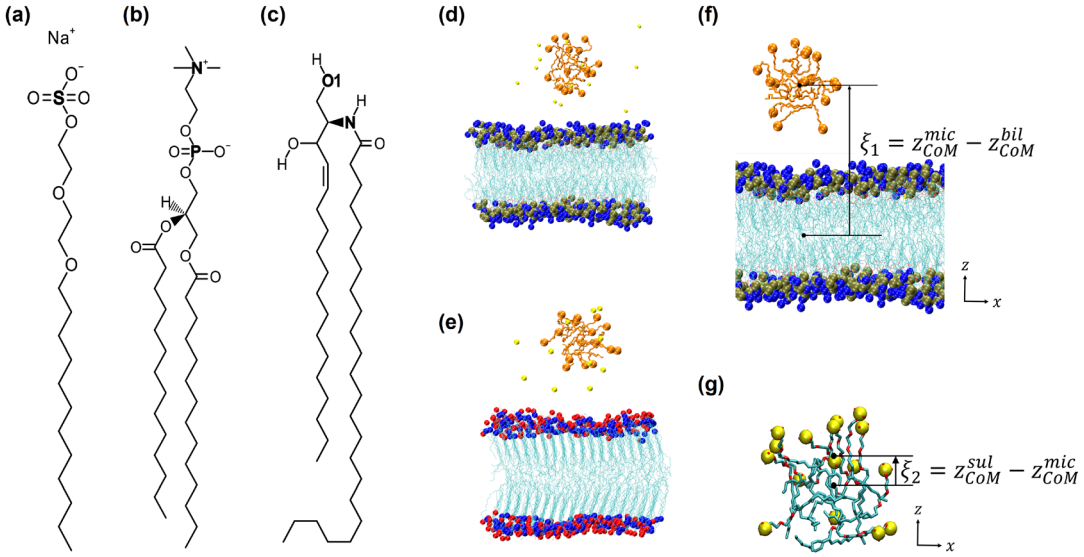


Fig. 1 Chemical structures of (a) the SLE2S surfactant, (b) DMPC, (c) Cer240 ceramide lipid molecules, respectively. Two simulation systems prepared to calculate 2D PMF profiles for a single SLE2S micelle placed on (d) a DMPC or on (e) a Cer240 lipid bilayer (lower). Definitions of two reaction coordinates, (e) ξ_1 and (f) ξ_2 for the 2D PMF constructions. For more detailed descriptions of the two reaction coordinates, see the text. In (d) through (f), the DMPC and Cer240 bilayer surfaces are normal to z direction and water molecules and hydrogen atoms are removed for clarity. For SLE2S micelles in (d) through (f), sulfur atoms are colored by orange spheres while other atoms are indicated by orange tubes. In addition, nitrogen (N), oxygen (O1), and phosphorus (P) atoms in lipid-heads are represented by blue, red, and tan spheres, respectively, and lipid tails are colored in cyan. In (g), sulfur atoms in SLE2S micelles are highlighted in big yellow spheres and oxygens and carbons are represented in red and cyan tubes, respectively. In (d) and (e), sodium cations are shown in yellow spheres.

has a polar head group, whereas Cer240 possesses a non-polar one. By constructing 2D potential of mean force (PMF) profiles based on reaction coordinates related to the position and shape of the micelle near the bilayer surface, we examined the influence of the head group polarity of cell membranes on micellar behavior. Additionally, to identify how surfactant transfer depends on the lipid head group polarity and the water molecules confined at the micelle–bilayer interface, 1D PMF curves were derived for surfactant transfer from the micelle to the two model bilayers.

2. Computational methods

2.1. Description of simulation systems

To examine the influence of the lipid head group polarity on micellar behavior and surfactant transfer, hydrated DMPC and Cer240 bilayer systems were separately established.[30,31] The SLE2S micelles, each composed of 16 surfactants, were adopted from our previous study.[18] Eight simulation systems were prepared, as summarized in Table 1. Systems 1 and 2 were designed to construct 2D potential of mean force (PMF) profiles to characterize the structural behavior of an SLE2S micelle near the DMPC and Cer240 bilayers (Fig. 1d and e). The construction of 2D PMF maps is described in the next subsection. A previous MD simulation study demonstrated surfactant transfer from an SLE2S micelle to the Cer240 bilayer.[18] To further investigate this surfactant transfer, 1D PMF calculations were performed in Systems 3–6 for a surfactant migrating from an SLE2S micelle to the DMPC and Cer240 bilayers at 313.15 K and 350.15 K. For Systems 7 and 8, constant-force steered MD (cf-SMD) simulations were conducted at 350.15 K in the DMPC bilayer and at 313.15 K in the Cer240 bilayer to verify the consistency of the

Table 1 Descriptions of eight MD simulation systems

	N							
System	SLE2S	DMPC	Cer240	Water	T (K)	P (bar)	Simulation time (ns)	21.6 cm Notes
1	16[ab]	200	—	23 000	313.15	1.0	15[c]	2D PMF
2	16	—	256	23 000	313.15	1.0	15[c]	2D PMF
3	16	200	—	23 000	313.15	1.0	40[d]	1D PMF
4	16	—	256	23 000	313.15	1.0	40[d]	1D PMF
5	16	200	—	23 000	350.15	1.0	30[d]	1D PMF
6	16	—	256	23 000	350.15	1.0	30[d]	1D PMF
7	16	200	—	23 000	350.15	1.0	400	cf-SMD
8	16	—	256	23 000	313.15	1.0	400	cf-SMD

[a] The number of surfactants in a micelle. [b] The same number of sodium cations were also included as counterions to neutralize the system. [c] Simulation time in each US window for 2D PMF. Last 5 ns was used for production run. [d] Simulation time in each US window for 1D PMF. Last 10 ns was used for production run.

obtained 1D PMF profiles with previous cf-SMD simulation results.[18] The SLE2S surfactant was modeled using the CHARMM general force field,[32,33] and the CHARMM36 force field set was used for the DMPC and Cer240 ceramide lipid molecules.[34,35] Water molecules were treated using the TIP3P model. All simulations were performed in the NPT ensemble with periodic boundary conditions in all three directions. Temperature was maintained at 313.15 K or 350.15 K using the Langevin thermostat with a damping coefficient of 3 ps^{-1}. The Nosé–Hoover Langevin piston method was used to semi-isotropically control pressure at 1 bar.[36] Electrostatic interactions were evaluated using the particle-mesh-Ewald (PME) summation, while van der Waals (vdW) interactions were modeled using the Lennard-Jones potential with cutoff and switching distances of 12 and 10 Å, respectively. MD simulations were performed with a time step of 2 fs *via* the NAMD 2.13 program package, and system visualization was conducted using the VMD 1.9 program software.[37,38]

2.2. 2D PMF constructions for energetics and structural behavior of an SLE2S micelle on two types of lipid bilayers

The 2D PMF profiles, $\Delta G(\xi_1, \xi_2)$, were obtained by combining MD simulations with the US method, with postprocessing performed using the weighted histogram analysis method (WHAM),[39,40]

$$\Delta G(\xi_1, \xi_2) = -\frac{1}{\beta}\ln\langle\delta(\xi_1 - \xi_1(\mathbf{R}))(\xi_2 - \xi_2(\mathbf{R}))\rangle. \quad (1)$$

To construct the 2D PMF surfaces, we introduced two reaction coordinates related to the micellar position and shape (Fig. 1f and g). The first reaction coordinate, ξ_1, represents the vertical distance between the micelle and bilayer, defined as

$$\xi_1 = z_{\mathrm{CoM}}^{\mathrm{mic}} - z_{\mathrm{CoM}}^{\mathrm{bil}} \quad (2)$$

where $z_{\mathrm{CoM}}^{\mathrm{mic}}$ is the z-coordinate of the center of mass (CoM) of an SLE2S micelle, and $z_{\mathrm{CoM}}^{\mathrm{bil}}$ is the average z-coordinate of phosphorus atoms in the DMPC bilayer or nitrogen atoms in the Cer240 bilayer head groups. The second reaction coordinate, ξ_2, characterizes the micellar deformation, defined as

$$\xi_2 = z_{\mathrm{CoM}}^{\mathrm{sul}} - z_{\mathrm{CoM}}^{\mathrm{mic}} \quad (3)$$

where $z_{\mathrm{CoM}}^{\mathrm{sul}}$ is the average z-coordinate of all sulfur atoms in the SLE2S micelle (Fig. 1g). To systematically vary ξ_2 during the US simulations, a harmonic biasing potential was applied to restrain ξ_2 to a specific position. A value of $\xi_2 \approx 0$ indicates a nearly spherical micelle. Values of ξ_2 above and below zero correspond to upper and lower hemispherical shapes, respectively. To validate ξ_2 as a reaction coordinate for micellar deformation, an additional PMF profile was computed as a function of ξ_2 through MD simulations using the US method for an SLE2S micelle in bulk water at 313.15 K. For the validation PMF calculations, a harmonic biasing potential was applied to restrain ξ_2 to a specific position with a force constant of 8.0 kcal mol^{-1} $Å^{-2}$. ξ_2 was varied from −4.0 Å to 4.0 Å with a window width of 0.5 Å. In addition, to maintain the structural integrity of the micelle, each SLE2S molecule was restrained within 17.0 Å of the micelle center using a flat-bottom half-harmonic potential with a force constant of 0.5 kcal mol^{-1} AA^{-2}. The resulting PMF profile (Fig. S1, ESI†) exhibited the energy minimum at $\xi_2 = 0$, as expected, confirming that ξ_2 effectively captures the energy variations associated with micellar deformation. For the 2D PMF profiles, ξ_1 was varied from 32.0 Å to 43.0 Å and from 34.0 Å to 47.0 Å for the micelle on the DMPC and Cer240 bilayers, respectively, with a window width of 0.5 Å. In contrast, at each ξ_1 position, ξ_2 was varied from −4.0 Å to 4.0 Å for the micelle on both bilayers, with a window width of 0.5 Å. With this configuration, the total number of sampling windows reached 357 and 459 for the DMPC and ceramide bilayers, respectively. The biasing force constants were set as 4.0 kcal mol^{-1} $Å^{-2}$ for ξ_1 and 8.0 kcal mol^{-1} $Å^{-2}$ for ξ_2 in both bilayer systems. Each window included a 10 ns equilibration run followed by a 5 ns production run.

2.3. 1D PMF calculations for surfactant transfer

1D PMF profiles were obtained to assess the transfer of an SLE2S surfactant from a micelle to a model DMPC or Cer240 lipid bilayer at two different temperatures (313.15 and 350.15 K) using the US method. For these calculations, we considered only the direct transfer of SLE2S surfactant from the micelle to the bilayers. The SLE2S micelle was restrained at its minimum energy point on each bilayer using the same force constants as in the 2D PMF constructions (Fig. 2a and b). When the micelle was restrained, its CoM was calculated using the fifteen remaining surfactant molecules, excluding the transferring molecule. A flat-bottom harmonic constraint with a weak spring constant of 0.5 kcal mol^{-1} $Å^{-2}$ was applied to all surfactants except the transitioning one, ensuring that the individual CoM of the remaining surfactants were restrained within a distance of 17.0 Å of their total CoM. Another flat-bottom harmonic constraint was applied to the transferring surfactant to restrain its lateral position within 20 Å and 30 Å of the micelle center for the DMPC and Cer240 bilayers, respectively. The umbrella biasing potential was applied to the CoM of the transferring surfactant using a force constant of 4.0 kcal mol^{-1} $Å^{-2}$ and bin width of 0.5 Å. The temperature dependence of surfactant transfer was examined by performing PMF calculations at 313.15 K and 350.15 K for both bilayer systems, assuming that the minimum energy points of the micelle remained unchanged between the two temperatures. Each window included 10 ns production run after 30 ns equilibration run at 313.15 K and 20 ns equilibration run at 350.15 K. The 1D PMF results were validated using cf-SMD simulations, performed at 350.15 K for the micelle-DMPC bilayer system and at 315.15 K for the micelle-Cer240 bilayer system. The cf-SMD simulations for the micelle-DMPC bilayer system at 313.15 K and the micelle-Cer240 bilayer system at 350.15 K had been conducted in our previous work.[18]

3. Results and discussion

The 2D PMF profiles as a function of the location and shape of an SLE2S micelle are presented in Fig. 2, demonstrating the

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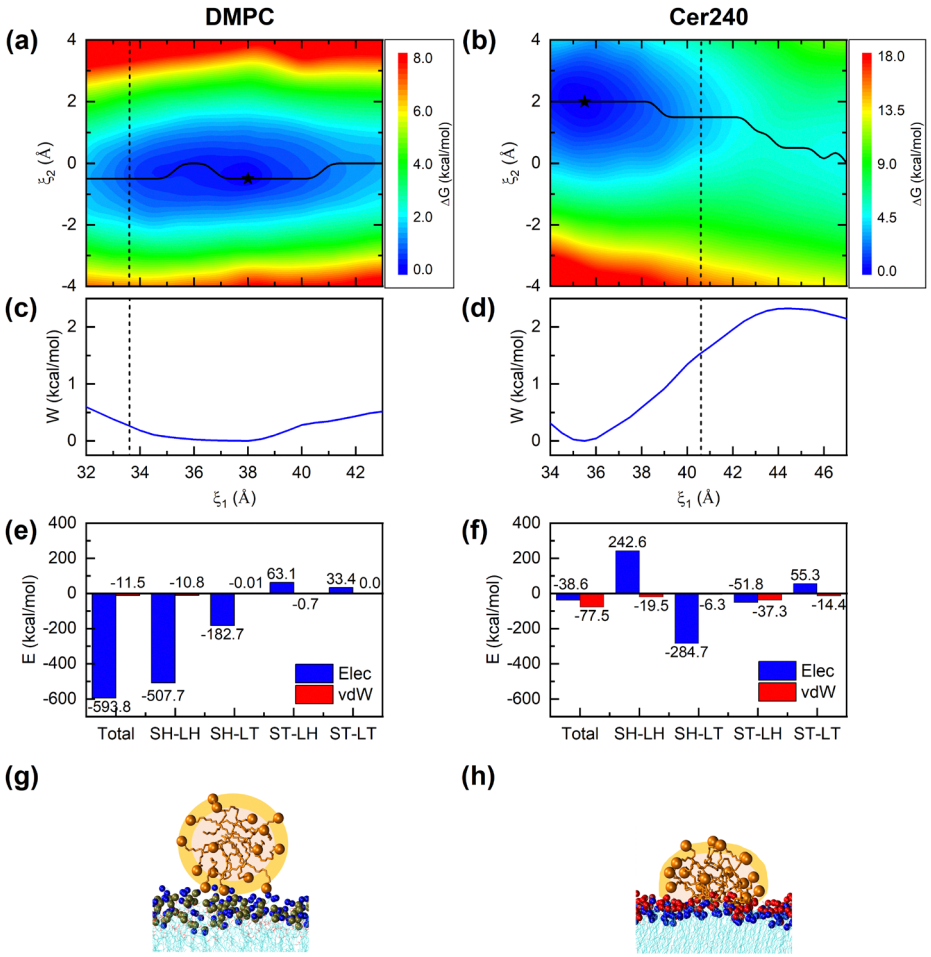


Fig. 2 (a) and (b) Two-dimensional (2D) and (c) and (d) one-dimensional (1D) PMF profiles for an SLE2S micelle approaching to the surface of DMPC and Cer240 bilayers as a function of both ξ_1 and ξ_2 ((a) and (b)) and ξ_1 only ((c) and (d)) at 313.15 K. (e) and (f) Electrostatic (Elec) and van der Waals (vdW) contributions to the interactions between the micelle surfactant heads (SHs) and tails (STs) and the lipid heads (LHs) and tails (LTs) of the bilayers at the global energy minima of ΔG that are indicated by the star symbols at (38.0 Å, −0.5 Å) for the DMPC bilayer and at (35.5 Å, 2.0 Å) for the Cer240 bilayer in the 2D PMF profiles in (a) and (b). (g) and (h) Shapes of the micelles at the global energy minima in the 2D PMF profiles in (a) and (b). Note that the range of ξ_1 and energy scale differ between (a) and (b). In (a) and (b), the solid black lines corresponds to the minimum energy pathways along the reaction coordinate, ξ_1 in the 2D PMF contour maps. The vertical dashed lines at z = 33.6 Å in (a) and (c) and at z = 40.6 Å in (b) and (d) represent the bilayer-micelle contact distance, d_{bmc} for the DMPC and Cer240 bilayers. In (e) and (f), the definitions of SH, ST, LH, and LT groups are illustrated in Fig. S4 of ESI.† In (g) and (h), the atom colors are the same as those in Fig. 1(d) and (e) and the thick yellow lines around the micelles were added for clarity to illustrate the distribution of surfactant head groups in the micelles.

energetic and structural differences of an SLE2S micelle on the DMPC and Cer240 bilayers. To facilitate analysis, we define the bilayer-micelle contact distance, d_{bmc}, as the distance in the z direction between the lipid bilayer center and the center of mass (CoM) of a perfectly spherical micelle in direct contact with the bilayer surface. Given that the half-thickness of the DMPC bilayer is approximately 18.0 Å and the radius of a spherical SLE2S micelle in bulk water is approximately 15.6 Å,[18,41] the contact distance is estimated to be d_{bmc}^{DMPC} = 33.6 Å. Similarly, for the Cer240 bilayer, which has a half-thickness of 25.0 Å,[42] the contact distance is d_{bmc}^{Cer240} = 40.6 Å. The distance parameters are indicated as vertical dashed lines in Fig. 2 through 4. Given d_{bmc}^{DMPC} = 33.6 Å, the minimum energy point at (ξ_1, ξ_2) = (38.0 Å, −0.5 Å) in Fig. 2a (also in Fig. S2, ESI†) indicates that the SLE2S micelle is remains nearly spherical and is suspended approximately 4–5 Å above the DMPC bilayer surface rather than being directly attached to it. In addition, the relatively broad energy well along the ξ_1 direction implies that the SLE2S micelle is not tightly bound at the energy minimum and instead drifts along the surface. In contrast, the narrow energy well around zero in the ξ_2 direction suggests that the micelle maintains its spherical shape near and even on the DMPC surface with minimal structural change (Fig. S2c, ESI†).

The deformation-related reaction coordinate, ξ_2 = −0.5 Å, at the minimum indicates that surfactants in the micelle are slightly more populated in the lower half of the spherical micelle near the DMPC bilayer, consistent with findings from a previous study.[18] On the other hand, the micelle on the Cer240 bilayer exhibits a deeper and narrower energy well located in the upper left region of the 2D PMF shown in Fig. 2b (also in Fig. S3, ESI†). Compared with that for the DMPC bilayer, the reaction coordinate, ξ_1 = 35.5 Å at the energy minimum, is considerably

smaller than the bilayer-micelle contact distance, d_{bmc}^{Cer240} = 40.6 Å on the Cer240 bilayer. Together with the deformation reaction coordinate, ξ_2 = 2.0 Å at the energy minimum, this observation indicates that the SLE2S micelle is tightly bound to the surface of the Cer240 bilayer and is even suppressed on the surface, leading to its deformation into an upper hemispherical shape (Fig. S3b and c, ESI†).[18] Additionally, a saddle point is observed at (ξ_1, ξ_2) = (43.5 Å, 1.0 Å) where the micelle begins to transition from a spherical to a slightly upper hemispherical shape with an increase of free energy in Fig. 2b (also in Fig. S3f, ESI†). To determine the position of the SLE2S micelle from the bilayer surface, 1D PMF curves, $W(\xi)$, were constructed as a function of ξ_1 by integrating the 2D PMF contour maps along the ξ_2 direction based on the equation:[43]

$$W(\xi_1) = -\frac{1}{\beta}\ln\int \exp(-\beta\Delta G(\xi_1, \xi_2))d\xi_2, \quad (4)$$

where β is the reciprocal of the product of the Boltzmann constant and temperature. In Fig. 2c, the shallow energy well, located beyond the micelle contact distance for the DMPC bilayer, confirms that the SLE2S micelle does not bind to the DMPC surface but instead hovers above it. In contrast, a deep energy minimum, well within the micelle contact distance for the Cer240 bilayer, indicates that the micelle strongly adheres to the Cer240 bilayer surface. The 1D PMF curve in Fig. 2d confirms the saddle point observed in the 2D PMF profile in Fig. 2b. This saddle point likely appears because of the rearrangement of several hydrophobic tail groups of the micelle, leading to their exposure to the water phase to interact with the non-polar Cer240 bilayer when the micelle approaches the bilayer surface. Electrostatic and vdW contributions to the interactions between the SLE2S micelle and both bilayers at the global energy minima are illustrated in Fig. 2e and f. In the case of DMPC bilayer, attractive electrostatic interactions between the charged surfactant head groups and polar lipid head groups are dominant and stabilize the SLE2S micelle at the energy minimum. In contrast, vdW interactions are extremely weak (Fig. 2e). Notably, these attractive interactions are attributable to the negatively charged sulfate groups in the SLE2S surfactants and positively charged choline groups of the DMPC lipid molecules. For the Cer240 bilayer, electrostatic interactions between the head and tail groups in the surfactant and lipid molecules largely cancel each other out, reducing their contribution compared with that in the case of the DMPC bilayer (Fig. 2f). Although vdW interactions become more dominant than electrostatic interactions, both interactions contribute at the lowest energy point. Because the key difference between the DMPC and Cer240 bilayers lies in the polarity of their lipid head groups, our observations suggest that the behavior of surfactant micelles, including micellar shape and surface adhesion, is primarily governed by this polarity through electrostatic and vdW interactions. Fig. 2g and h illustrate typical SLE2S micellar shapes at the energy minimum point on each bilayer. The micelle maintains a distance from the DMPC bilayer surface and retains its spherical shape (Fig. 2g), whereas it adheres tightly to the Cer240 bilayer surface, adopting an upper hemispherical geometry (Fig. 2h). To examine the effect of the SLE2S micelle on the surfaces of DMPC and Cer240 bilayers, we calculated the change in the local thickness of the upper leaflet of the bilayer as the micelle approaches the bilayer surface. The local thickness is defined as the thickness of the upper leaflet composed of lipid molecules whose CoMs lie within a radius of l from the CoM of the micelle in the lateral direction (Fig. S5, ESI†). This thickness is measured as the average distance between the bilayer center and P atoms in

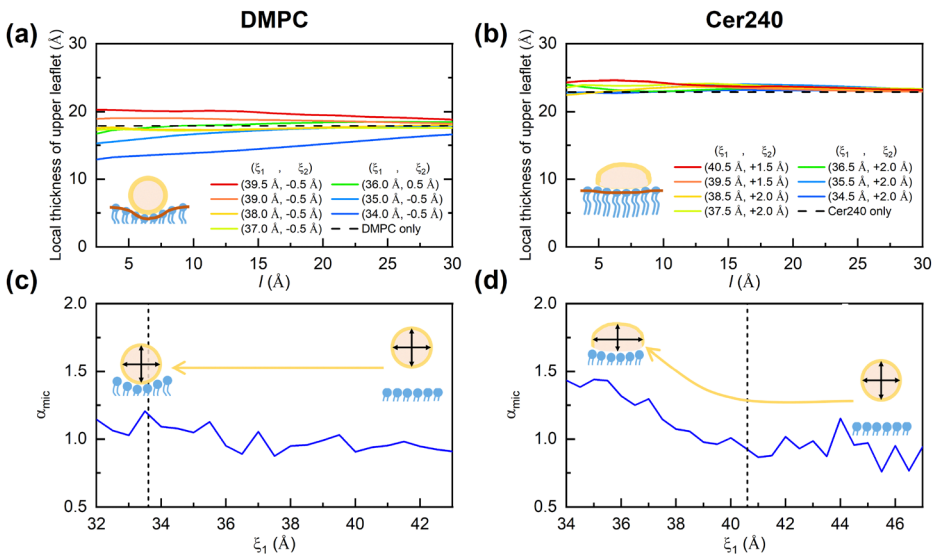


Fig. 3 The local thickness of the bilayer upper leaflet for (a) the DMPC bilayer and (b) the Cer240 bilayer at the several locations that are selected from the minimum energy pathways as shown in the 2D PMF profiles of Fig. 2a and b. The average aspect ratios of the micelle as the micelle approaches to (c) the DMPC and (d) the Cer240 bilayer along the minimum energy pathways. In (a) and (b), the horizontal broken lines represent the average thickness of the upper leaflet in the DMPC and Cer240 bilayers in the absence of the micelle. The vertical dashed lines in (c) and (d) are the bilayer-micelle contact distance, d_{bmc} for each bilayer as shown in Fig. 2a–d.

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the DMPC or N atoms in the Cer240 bilayer. Fig. 3a illustrates that when the SLE2S micelle is far from the DMPC bilayer surface (*i.e.*, at (ξ_1, ξ_2) = (39.5 Å, −0.5 Å)), the local thickness slightly increases to have favorable interactions with the micelle. However, as the micelle approaches the surface, the thickness gradually decreases, and the bilayer surface becomes concave. This significant change in curvature indirectly implies that the DMPC bilayer exists in a flexible liquid crystal (L_c) fluid phase at the specified temperature. Moreover, the concave surface indicates repulsive interactions when the SLE2S micelle approaches the DMPC bilayer surface too closely. In contrast, for the Cer240 bilayer surface (Fig. 3b), the local thickness of the upper leaflet remains largely unchanged as the micelle approaches the surface although a slight increase in the thickness is observed when the micelle is distant from the surface. The minimal change in the thickness of the Cer240 bilayer is attributed to its rigid structure in the gel phase at the specified temperature.[42] The influence of interactions with the bilayers on micellar shape was assessed by calculating the aspect ratio of the micelle, defined as

$$\alpha_{\mathrm{mic}} = \frac{2\langle I_z \rangle}{\langle I_x \rangle + \langle I_y \rangle} \quad (5)$$

where I_x, I_y, and I_z denote the components of the principal moment of inertia.[44] Fig. 3c shows that the aspect ratio of the SLE2S micelle remains nearly constant irrespective of its distance from the DMPC bilayer surface. In contrast, the ratio increases substantially as the micelle approaches the Cer240 bilayer (Fig. 3d), supporting the observation that the micelle deforms from a spherical to an upper hemispherical shape near the Cer240 bilayer surface. The aspect ratio behavior in both bilayers is consistent with the 2D PMF results (Fig. 2a and b), as well as prior calculations of the radius of gyration calculations for the SLE2S micelle on the two bilayers.[18]

Our previous MD simulation study showed that several surfactant molecules transferred from the SLE2S micelle to the Cer240 bilayer at 350.15 K, whereas no such transfer was observed in the DMPC bilayer at 313.15 K.[18] To explore this phenomenon, we calculated the free energy changes associated with surfactant transfer at 313.15 and 350.15 K, as illustrated in Fig. 4a and b. In these PMF calculations, we assumed that the energy minima of the micelle on the DMPC and Cer240 bilayers at 350.15 K remained the same as those at 313.15 K (Fig. 2a and b). For the surfactant migrating from the micelle to the DMPC bilayer, the PMF profiles at both temperatures are nearly identical and show energy barriers of approximately 1.8 kcal mol^{-1}. These barriers are sufficiently high to prevent surfactant transfer under both temperatures. Moreover, the barrier heights are consistent with the free energy increase observed in our previous study where an SLE2S surfactant was displaced from its energy minimum within the micelle to a position 5 Å away in bulk water.[18] where the free energy difference between the minimum and a point 5 Å away The PMF predictions are further supported by 400 ns constant-force steered MD (cf-SMD) simulations (Fig. S6a and b, ESI†), which confirm the

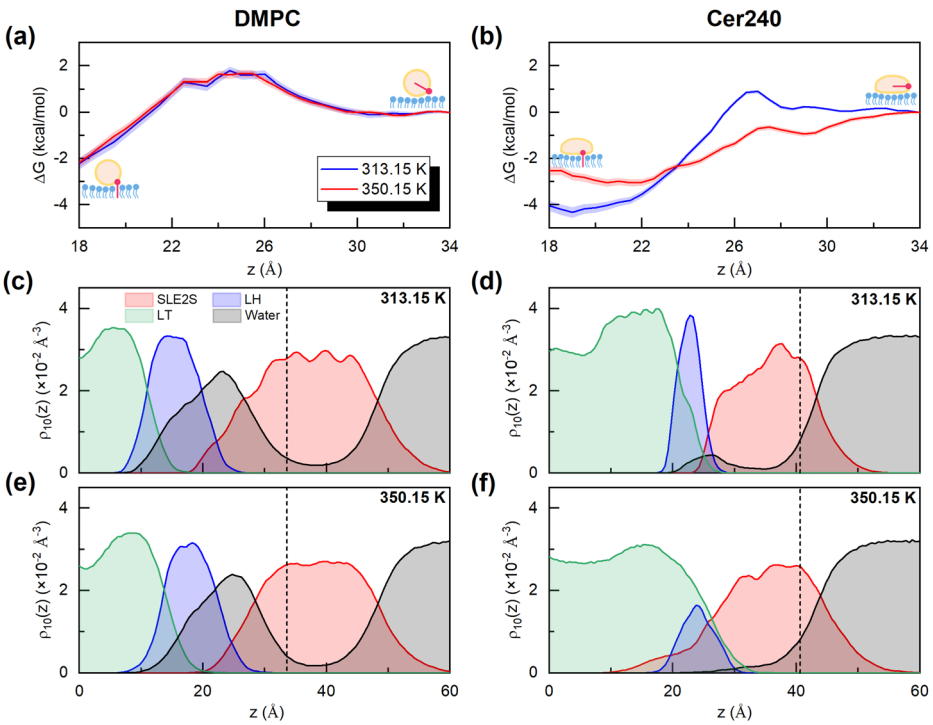


Fig. 4 Free energy changes of a surfactant transfer from the SLE2S micelle to the (a) DMPC and (b) Cer240 bilayers at the temperature of 313.15 K and 350.15 K and the number density distributions for the SLE2S surfactants, lipid head (LH) and lipid tail (LT) groups, and water molecules contained in the cylinder with a lateral radius of l = 10 Å at (c) 313.15 K and (e) 350.15 K in the DMPC bilayer and at (d) 313.15 K and (f) 350.15 K in the Cer240 bilayer as a function of distance z from the bilayer center. The definition of the cylinder with a lateral radius of l is presented in Fig. S5 of ESI.† The vertical dashed lines in (c) through (f) represent the bilayer-micelle contact distance, d_{bmc} for each bilayer as shown in Fig. 2a–d.

absence of surfactant transfer. In contrast, surfactant transfer to the Cer240 bilayer depends on the temperature (Fig. 4b). At 313.15 K, the energy barrier of 0.9 kcal mol^{-1} inhibits or delays surfactant transfer. However, this energy barrier disappears at 350.15 K. The temperature dependence of surfactant transfer in the Cer240 bilayer is further supported by the 400 ns cf-SMD simulations (Fig. S6c and d, ESI†), which show multiple surfactant transfer processes only at 350.15 K. To better understand how the SLE2S surfactant transfer depends on the lipid species in the bilayer and temperature, we calculated the number density profiles of the system components as a function of distance z from the bilayer center. These calculations considered only the components within a cylindrical region of lateral radius l = 10 Å centered at the CoM of the micelle (Fig. S5, ESI†). Fig. 4c and e show that a significant number of water molecules remain confined between the SLE2S micelle and DMPC bilayer at both 313.15 and 350.15 K. Our previous study revealed that surfactant transfer from the SLE2S micelle to bulk water requires an energy cost of approximately 5.6 kcal mol^{-1}.[18] As the SLE2S micelle remains distant from the DMPC bilayer, a transferring surfactant is partially exposed to the interfacial water phase, resulting in an energy barrier of 1.8 kcal mol^{-1} in the PMF curves (Fig. 4a). In contrast, significantly fewer water molecules are present between the micelle and Cer240 bilayer (Fig. 4d and f) as the SLE2S micelle adheres tightly to the bilayer (Fig. 2f). Moreover, Fig. 4d and f reveal a striking difference in the distributions of Cer240 head groups at 313.15 K and 350.15 K. At the lower temperature, these head groups are densely populated between the micelle and the lipid tail groups, forming an energy barrier that hinders surfactant transfer. As the temperature increases, the head groups become more loosely arranged, creating open spaces among lipid head groups that allow surfactants to enter the bilayer. This temperature-dependent reduction in head group density is further supported by area per lipid (APL) calculations for the pure Cer240 bilayer, which show an increase from 42.8 $Å^2$ at 313.15 K to 51.7 $Å^2$ at 350.15 K (Table S1, ESI†). In addition, in the case of the Cer240 bilayer at 350.15 K, a small opening was observed in the head group region directly beneath the micelle, which appeared to facilitate surfactant transfer (Fig. S7, ESI†).

4. Summary

Through MD simulations combined with free energy calculations, this study explores the influence of the polarity of lipid head groups on the behavior of negatively charged SLE2S micelles near membrane surfaces and the transfer of surfactants from the micelle to cell membranes. The 2D PMF profiles illustrate that the SLE2S micelle is repelled by the polar DMPC bilayer surface and maintains a nearly spherical shape. In contrast, it adheres tightly to the non-poplar Cer240 bilayer, adopting an upper hemispherical conformation. As expected, electrostatic interactions between the anionic SLE2S head groups and the polar DMPC head groups dominate at the minimum energy point in the 2D PMF profile. In contrast, in the case of the non-polar Cer240 bilayer, both electrostatic and vdW interactions contribute at the energy minimum although vdW interactions play a more dominant role. The 1D PMF calculations indicate that surfactant transfer from the SLE2S micelle to the DMPC bilayer is unlikely to occur at either 313.15 K or 350.15 K due to an energy barrier of 1.8 kcal mol^{-1}. This energy barrier arises from interfacial water molecules confined between the micelle and DMPC bilayer. In contrast, surfactant transfer occurs at 350.15 K in the Cer240 bilayer, as the head groups are more loosely arranged at higher temperatures, facilitating the transfer process. These 1D PMF predictions are supported by 400 ns cf-SMD simulations at both temperatures.

Overall, this study provides valuable insight into the mechanisms by which charged DDSs interact with and penetrate cell membranes, offering guidance for the development of novel DDSs based on surfactant micelles. Additionally, our findings can help characterize the role of surfactants as penetration enhancers in transdermal DDSs.

Author contributions

YS carried out the MD simulations. Data analysis and manuscript preparation were done by YS and HH.

Conflicts of interest

There are no conflicts to declare.

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